Lumina Strategies: System Architecture

1. High-Level Overview

Lumina is a serverless, decoupled quantitative finance dashboard. The front end is a React-based single-page application (SPA) that reads heavily from an AWS S3 data lake. Heavy computational lifting, API data fetching, and AI synthesis are handled by automated AWS container tasks and Lambda functions.

2. Frontend Application (React / Vite)

- **Hosting:** Vercel (CI/CD via GitHub).
- **State & Styling:** React Hooks, Tailwind CSS for premium dark-mode UI.
- **Core Modules:**
    - **Portfolio Tracker & Screener:** Screen 1300 stocks in Canadian and American markets. Reads parsed CSVs to track real-time closing prices, compute 1-12 month historical returns, and scan for algorithmic anomalies (e.g., deep value, volume spikes).
    - **Target Analysis Engine:** Projects current closing prices against historical analyst mean targets to calculate deep discount windows and highlight entry/exit points.
    - **Global Macro Correlator:** A dual-axis correlation engine aligning stock price action with 120+ economic indicators, featuring dynamic time-lag shifting to identify leading indicators and real-time Pearson Correlation math.
    - **AI News Engine:** Renders complex AI JSON outputs into interactive "Delta Matrices" comparing live news to stored S3 baselines.

3. Serverless Backend & APIs (AWS Lambda)

- **API Architecture:** Configured via direct Lambda Function URLs with restrictive CORS headers.
- **On-Demand Functions (Python 3.9):**
    - `lumina-watchlist-sync`: Manages user portfolio state and saves configurations directly to S3.
    - `lumina-macro-favorites`: Schema-agnostic bridge to save and retrieve specific stock/metric/lag/timeframe combinations to the dashboard tray.
    - `lumina-ai-research`: Orchestrates AI agents to scrape live financial news, compare it to historical S3 baseline theses, and output an assessed "Delta Matrix."

4. Automated Data Pipelines (AWS ECS Fargate & EventBridge)

- **Compute:** Dockerized Python scripts running on AWS ECS Fargate, fully abstracted from underlying EC2 infrastructure.
- **Scheduling:** AWS EventBridge triggers tasks automatically.
- **Data Collection Agents:**
  - **Market Data Collector (Fargate):** Pulls historical and current institutional targets for global indices (S&P 500, TSX, etc.) daily.
  - **Macro Data Collector (Fargate):** Queries the FRED API for 120+ macroeconomic indicators. Dynamically checks S3 to find the oldest portfolio stock date, factoring in an additional 360 days for delayed time-lag math.
  - **AI Sentiment Engine (Lambda):** A scheduled Lambda that scrapes CNBC RSS feeds daily and uses Google Gemini to evaluate overall market mood, appending it to a 30-day rolling momentum array.

5. Data Lake (AWS S3)

- **Bucket:** `lumina-strategies` (Strict bucket policies limiting `GetObject` access to specific dashboard data folders).
- **Structure:**
  - `/data/today/`: Daily overwrites of index stock targets and historical prices (CSV).
  - `/data/macro/`: Global macroeconomic indicator datasets (CSV).
  - `/data/historical-archive/`: Immutable daily `.csv` backups.
  - `/dashboard/watchlist/`: User portfolio configurations (JSON).
  - `/dashboard/favorites/`: Saved macro-correlation plays (JSON).
  - `/dashboard/research/`: Stored AI-generated thesis baselines (JSON).
  - `/dashboard/sentiment/`: 30-day rolling AI market mood (JSON).